



Evergreen Market Inventory Optimization

Data Analytics Case Study

Executive Summary

Evergreen Market is a nationwide retail chain with 350+ stores and a rapidly growing online presence. The company faces significant inventory challenges: overstocking low-demand items, stockouts of high-demand products, and inconsistent forecasting. These issues increase operational costs, reduce customer satisfaction, and weaken profitability.

This project analyzes Evergreen's 73,100-row retail dataset to uncover demand patterns, identify inefficiencies, and propose data-driven strategies for improving inventory alignment across regions and product categories.

Company Background

Evergreen Market offers pantry staples, personal care items, seasonal merchandise, and home essentials. Recent operational reports show:

- Overstocking → higher warehousing costs, markdowns, waste
- Stockouts → lost sales, customer dissatisfaction
- Reactive analytics → poor forecasting accuracy

Leadership is seeking a shift toward ***predictive, proactive inventory management***.

Dataset Overview

Rows: 73,100 Variables: 15 Includes:

- Inventory levels
- Units sold
- Units ordered
- Pricing & discounts
- Demand forecasts
- Product categories
- Regions
- Order dates

Data Quality Findings

- *715 outliers* in Units Sold
- *732 outliers* in Demand Forecast
- Skewed distributions
- Weak correlations
- High variability across categories & regions

These issues limit forecasting accuracy and highlight the need for improved data collection and modeling.

Data Needs Assessment

The current dataset captures transactional metrics but lacks external and behavioral data needed for accurate forecasting. Evergreen should incorporate:

- Customer purchasing patterns
- Regional events
- Online browsing behavior
- Weather or seasonal indicators

These additional inputs would explain demand fluctuations that the existing dataset cannot capture.

Methods & Analysis

1. Data Cleaning

- Verified missing values (none found)
- Confirmed zero duplicate records
- Detected outliers using IQR

2. Exploratory Data Analysis (EDA)

Visualizations created:

- Histograms
- Boxplots
- Violin plots
- Scatter plots (Units Sold vs. Demand Forecast)

- Correlation heatmap
- Pivot-based heatmap (Avg Units Sold by Category & Region)

Key EDA Insights:

- Demand forecasts often fail to match actual sales
- Category-region combinations show strong performance clusters
- Outliers distort forecasting accuracy
- Weak correlations indicate limited predictive power in current inputs

3. Modeling Approach

Future modeling should include:

- Regression analysis
- Clustering stores by demand profile
- Anomaly detection
- Forecasting models (ARIMA, Random Forest, Gradient Boosting)

Business Insights

- Evergreen's forecasting inputs lack strong predictive power
- Inventory misalignment is driven by reactive analytics
- High-performing category-region combinations deserve increased promotional investment
- Outliers and variability limit the reliability of current demand forecasts
- Predictive analytics would significantly reduce operational waste

Recommendations

Tool Recommendations

- Python + Scikit-Learn for predictive modeling
- Tableau or Power BI for interactive dashboards and KPI monitoring

Strategic Recommendations

- Adopt advanced predictive modeling tools
- Implement real-time dashboards

- Expand data collection to include behavioral and external variables
- Prioritize data cleaning workflows
- Redirect promotional investments to high-performing category-region pairs
- Develop anomaly detection systems
- Establish a continuous analytics improvement cycle

Personal Learning Reflection

This project strengthened my skills in:

- Python
- Data cleaning
- Visualization
- Exploratory Data Analysis
- Model evaluation
- Communicating insights to stakeholders

I gained hands-on experience with violin plots, pivot heatmaps, correlation heatmaps, and other advanced visualizations that will support my future work as a data analyst.